

Module 2: Data Preprocessing & Normalization

Data Cleaning, Chi-Square Correlation, Min-Max, Z-Score & PCA Reduction

Module II: Data Preprocessing — Complete 6-Topic Syllabus Tracker

Topic 15: Data Cleaning (Handling Missing Values & Noisy Data Smoothing)

Topic 17: Data Transformation (Min-Max & Z-Score Normalizations)

Topic 19: Data Discretization (Unsupervised Binning & Supervised Trees)

Topic 16: Data Integration (Schema Matching & Redundancy Correlation)

Topic 18: Data Reduction (Dimensionality, Numerosity & Compression)

Topic 20: Concept Hierarchy Generation for Nominal & Numeric Data

Memory Hook: C-I-T-R-D-H Preprocessing Pipeline

Clean → Integrate → Transform → Reduce → Discretize → Hierarchy

Topic 15: Data Cleaning Techniques

Missing Data Strategy	Operational Mechanism	Tradeoffs & Constraints
1. Ignore Tuple	Discard the entire row containing missing values.	Simple; highly wasteful if tuple contains other valuable attributes.
2. Global Constant	Fill missing values with a label like "Unknown" or $-\infty$.	May mislead data mining classifiers into treating "Unknown" as a true category.
3. Attribute Mean / Median	Replace missing numeric value with mean (symmetric distribution) or median (skewed data).	Preserves overall sample mean; reduces natural variance.
4. Most Probable Value	Estimate using regression, decision tree inference, or Bayesian expectation maximization.	Statistically soundest; computationally intensive.

Topic 17: Data Transformation & Normalization [UPLOADED PYQ]

1. Min-Max Normalization Formula [UPLOADED PYQ]:

$$v' = \frac{v - \min_A}{\max_A - \min_A}(\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

Maps values into designated target range $[\text{new_min}_A, \text{new_max}_A]$ (commonly $[0.0, 1.0]$).

2. Z-Score (Zero-Mean) Normalization Formula:

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

Where $\bar{A}$ is the mean and σ_A is the standard deviation. Essential when actual minimum and maximum are unknown or outliers dominate.

[UPLOADED PYQ] Solved Problem: Min-Max Normalization of Value 400 in Range [200, 1000] to Target [10, 20]

Given: $v = 400$, $\min = 200$, $\max = 1000$, $\text{new_min} = 10$, $\text{new_max} = 20$.

$$v' = \frac{400 - 200}{1000 - 200} \times (20 - 10) + 10 = \frac{200}{800} \times 10 + 10 = 0.25 \times 10 + 10 = 2.5 + 10 = \mathbf{12.5}$$

Topic 18 & 19: Data Reduction & Discretization

Reduction Approach	Core Mathematical / Statistical Techniques	Key Benefits
1. Dimensionality Reduction	Principal Component Analysis (PCA): Projects d -dimensional space onto k orthogonal eigenvectors ($k \ll d$) maximizing variance. Wavelet transforms.	Removes irrelevant features; mitigates Curse of Dimensionality.
2. Numerosity Reduction	Parametric: Linear regression models ($y = mx + c$), Log-linear models. Non-parametric: Histograms, Clustering, Random Sampling without replacement.	Replaces massive raw dataset with compact mathematical surrogate models.
3. Discretization (Binning)	• Equal-Width (Distance) Partitioning: Divides range into k equal intervals: $W = \frac{\max - \min}{k}$. • Equal-Frequency (Depth) Partitioning: Divides sorted values such that each bin contains N/k samples.	Converts continuous attributes into discrete intervals for Decision Trees and Association Mining.

Topic 20: Concept Hierarchy Generation [UPLOADED PYQ]

[UPLOADED PYQ – IT426 End] Concept Hierarchy Generation for Nominal Data

A **Concept Hierarchy** defines a sequence of mappings from a set of low-level concepts to higher-level, more general concepts:

Street $\longrightarrow$ City $\longrightarrow$ State $\longrightarrow$ Country

Two Primary Methods of Generation:

1. **Specification of Partial Ordering by Users / Experts at Schema Level:** Explicitly defined by database administrators (e.g., 'street < city < state < country').
2. **Data-Driven Automatic Generation Based on Attribute Value Counts (Distinct Cardinality):** The attribute with the largest number of distinct values is placed at the lowest level of the hierarchy, while the attribute with the fewest distinct values is placed at the top (e.g., 'Country' (10 values) > 'State' (50 values) > 'City' (500 values) > 'Street' (50,000 values)).

M2 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Explain Equal-Width vs. Equal-Frequency Binning for data smoothing on sorted data: `4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34`. (8 Marks)

Dataset size $N = 12$, target bins $k = 3$:

– **Equal-Frequency (Depth) Partitioning (4 elements per bin):**

- Bin 1: `4, 8, 9, 15` → Smoothed by bin mean: `9, 9, 9, 9`
- Bin 2: `21, 21, 24, 25` → Smoothed by bin mean: `23, 23, 23, 23`
- Bin 3: `26, 28, 29, 34` → Smoothed by bin mean: `29, 29, 29, 29`

– **Equal-Width Partitioning:** Width $W = \frac{34-4}{3} = 10$.

- Bin 1 $[4, 14)$: `4, 8, 9`
- Bin 2 $[14, 24)$: `15, 21, 21`
- Bin 3 $[24, 34]$: `24, 25, 26, 28, 29, 34`